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Next-gen plant clonal ecology

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ABSTRACT

Plants with clonal growth can produce multiple potentially independent units, termed ramets. Clonal growth can have important ecological and evolutionary consequences, such as by increasing probability of reproduction, space monopolization, and regeneration after injury; and by permitting physiological integration of connected ramets. Although clonal growth is widespread among species and habitats, it has received relatively little attention in plant ecology. To introduce this special issue on clonal plant ecology, we first provide a brief background on the topic, noting its importance in areas ranging from evolution to the impacts of climate change on plant communities. We then focus on a set of pressing questions, to highlight both the obstacles and opportunities to more explicitly incorporate clonal growth in research on plant ecology and evolution.

1. Purpose*

Clonal growth in plants is the ability to produce multiple potentially physiologically independent units, termed ramets, of the same genetic individual, or genet, via vegetative growth of stems, roots, or leaves. Although about two-thirds of plant species are capable of clonal growth (Herben and Klimešová, 2020), clonality is often overlooked in plant research (but see Larson and Funk, 2016; Salguero-Gómez, 2018) due to factors such as unfamiliarity with the differential responses of clonal and non-clonal plants, lack of attention to plant architecture and morphology, and the difficulty of studying the belowground organs responsible for much clonal growth. Spreading knowledge about the role of clonality in plant ecology is therefore crucial.

The importance of clonal growth in plant ecology and evolution is illustrated by the breadth of the 17 contributions to this special issue on the ecology and evolution of clonal growth. These papers cover trans-generational effects (Dong et al., 2019a*), species interactions (Wang et al., 2019*; Bittebiere et al., 2020*; Duchoslavová and Herben, 2020*), invasion biology (Roiloa, 2019*; Wang et al., 2019*), life history (Janovský and Herben, 2020*), allocation (Goldberg et al., 2020*), biodiversity conservation (Amor et al., 2020*), population structure (Ricono et al., 2020*), and responses to environmental stress (Huebner et al., 2019*) and disturbance (Huebner et al., 2019*; Vesk and Yen, 2019*; Escandón et al., 2020*; Franklin et al., 2020*; Martínková et al., 2020*). Understanding of clonal plant ecology is needed to adequately model plant community assembly and ecosystem function (Bittebiere et al., 2020*) and responses to climate change (Chelli et al., 2020*; Lubbe and Henry, 2019*; Watts et al., 2019*) and environmental heterogeneity (Franklin et al., 2020*). To introduce the special issue, we begin with a brief review of previous work on clonal plants and then consider some key current questions and challenges.

2. Background

In contrast to most animals, nearly all plants are modular (Harper and Bell, 1979): they maintain meristems with indeterminate growth that produce new shoot and root modules which add to or replace old ones (Ottaviani et al., 2017). In many plants, sets of modules can become physiologically independent. How clonality affects plant architecture has been stressed many times (see review by Oborny, 2019). However,

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* Asterisks indicate papers in the special issue on clonal plants.

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the ecological and evolutionary consequences of clonal growth have not received sufficient attention. Clonal growth can provide reproductive insurance when sexual reproduction is not possible (Herben et al., 2015); facilitate monopolization of space (Zheng et al., 2019), regeneration after injury (Martínková et al., 2020*), and vegetative dormancy (Shefferson, 2009; Ott et al., 2019); and permit physiological integration between connected ramets (Qureshi and Spanner, 1971, 1973;Alpert and Stuefer, 1997; Jaafry et al., 2016; Liu et al., 2016) and even potential immortality of genotypes (Eriksson, 1993; Klimešová et al., 2015). Since some of these properties are lacking in species that reproduce asexually only via vagile propagules such as apomictic seeds, bulbils, or plantlets, the term clonal growth, and its synonym clonality, are commonly used to refer only to asexual reproduction via non-vagile propagules.

Clonality is widespread in the plant kingdom, but it is by no means universal. It is more common in herbaceous species than in woody species (Aarssen, 2008). A recent phylogenetic analysis found that clonality, while showing distinct phylogenetic patterns in angiosperms, has been repeatedly evolved and lost (Herben and Klimešová, 2020), implying that selective pressures can favor or disfavor clonality. This suggests that clonality has advantages but also costs, and that these are likely to be determined by the biotic and abiotic environment.

Clonal growth plays a role in the ecology of some economically important plant groups. Some important crops are clonal, such as banana, potato, sweet potato, and manioc (Denham et al., 2020; McKey et al., 2010). Clonal growth has long been explored as a means to quickly increase world production of key crop species (Allaby, 2019; Denham et al., 2020), albeit with the risk of reduced genetic diversity (Keneni et al., 2012). Clonal plants, especially those with extensive systems of rhizomes or roots, can be important in the prevention of soil erosion (Guerrero-Campo et al., 2008). One economic cost of clonality may be promotion of biological invasions; clonal species are disproportionately represented among introduced, invasive plants in some floras (Pyšek et al., 2015).

Existing data show that different forms of clonal growth are associated with key environmental factors, including disturbance and stress. Different types of disturbance respectively favor clonal and non-clonal plants (Clarke et al., 2015; Bellingham and Sparrow, 2000), mainly due to regeneration potential and bud banks. Prevalance of clonality is affected by water availability (Ye et al., 2014; Klimešová et al., 2016), possibly due to aeration potentials of clonal growth organs or to ease of production of new roots (Sosnová et al., 2010). Proportions of different clonal growth forms (e.g., root- or stem-based) and of clonal plants overall change along environmental gradients, often nonlinearly (Herben et al., 2018).

Clonality in plants can be attained by several morphological means. New ramets may be produced along stems, roots, or even leaves (Klimeš et al., 1997) and clonal species differ in bud bank and other clonal growth traits (Klimešová et al., 2019). Experimental data clearly link these traits to the functioning of clonal plants (Liu et al., 2016; Martínková et al., 2020*). For example, plants with different clonal growth organs differ in the ability to share resources between connected ramets, which can strongly affect clonal fitness (Martínková et al., 2020*). We also know that clonal plants forage for nutrients by roots in heterogenous soil less than non-clonal counterparts (Weiser et al., 2016); such differences may help explain coexistence of different growth strategies in one community (Vojtkó et al., 2017).

Clonal growth can affect evolution in a variety of ways. Clonality tends to increase generation time, which can save costs of meiosis but slow adaptation (Vallejo-Marín et al., 2010; Gibson et al., 2017; Salguero-Gómez, 2018). Number of meiotic divisions is also hypothesized to be an important driver of non-adaptive molecular evolution in plants (Lanfear, 2013). Some types of clonality may confer a direct microevolutionary advantage in some contexts by reducing probabilities of speciation or extinction (Figueir et al., 2019). Different types of clonality are associated with different reproductive systems, generation times, and dispersal strategies, all of which influence rates of diversification (e.g., Roquet, 2013; Van Drunen and Husband, 2019).

Much of the literature on clonal plants has been produced by two informal groups of researchers, one centered in Europe and a newer one centered in China. Research presented in association with the conferences held by these groups over the past several decades has been published as books and special issues of journals (e.g.,de Kroon and van Groenendael, 1995; Dong et al., 2014; Gross et al., 2017). We are pleased to continue this tradition with this special issue based on "Clones in Context", the international conference on clonal plants held in August 2018 at Bowdoin College in Brunswick, Maine, USA, and on a combined special issue on clonal ecology.

3. Moving forward with clonal ecology – pressing questions

3.1. What can population genetics tell us about clonal plant ecology?

3.1.1. What we know

Clonal plants call for an added dimension of analysis that can capture the horizontal and temporal distribution of genotypes across the landscape. As extremely modular organisms, clonal plants, whether large integrated clones or fragmented into numerous independent units, challenge our notions of individuality and population genetics. Ultimately, the genetic footprint of a clonal genotype can be considerably broader than that of a non-clonal genotype. Replicated across the landscape, a large genotype can have a multiplier effect on traits such as fecundity. The potential longevity of clonal plants strains temporal concepts such as cohorts and generations. Genetic individuals can range over hundreds of kilometers and live for thousands of years. Each additional ramet is potentially an added source of pollen and seeds; thus, fitness is likely to increase with the spatial and temporal extent of the genotype (de Witte and Stöcklin, 2010). Special care is required when applying fundamental non-clonal theory and models such as the Hardy-Weinburg Equilibrium to clonal systems. Cautious interpretation is required for population genetic metrics such as heterozygosity or genetic diversity that could impact scientific, management, and conservation efforts, as these values will vary greatly depending on the extent of clonality and the definition of the individual (Douhovnikoff and Leventhal, 2016).

Due to less frequent sexual reproduction and potential interbreeding across multiple generations, it has been hypothesized that clonal plant populations should have lower genetic diversity than non-clonal plant populations and that genetic diversity may be lost over time as thinning reduces the population to a few large individuals. However, there are few examples of genetic erosion. In fact, observations of higher than anticipated diversity are not unusual and may be due to the great longevity of genotypes (de Witte and Stöcklin, 2010) as well as microsite heterogeneity. Also, many clonal plant species are polyploid, and the additional sets of chromosomes could potentially balance relatively low levels of inter-genet diversity with relatively high levels of intra-genet diversity. In effect, each individual has a deeper pool of genetic diversity to draw upon.

Recent work has also indicated that epigenetic variation may play a role in compensating for possible reduced reliance on sexual recombination among clonal genotypes (Dong et al., 2019b). Epigenetic mechanisms can allow for the acclimation of optimal phenotypes and the transfer of those optimizations to daughter ramets. In this way, heritable acclimation can supplement adaptation (Douhovnikoff and Dodd, 2015; González et al., 2018). For example, Dong et al. (2019b) point out that epigenetic regulation may be an explanation for the transgenerational effects they observed to be transmitted to non-integrated offspring ramets of *Alternanthera philoxeroides*.

3.1.2. Research needs

Our exploration of the extent and role of clonality in plant population genetics is still at an early stage. More species are now recognized as clonal (e.g., Herben and Klimešová, 2020), but, as observed by several authors in this issue, our ability to predict clonal structure and to understand the variables that influence clonal diversity is still limited (Huebner et al., 2019*; Watts et al., 2019*; Ricono et al., 2020*). These studies are good examples of the work that is necessary to better understand the influence of clonality on population genetics and on evolution of clonal species. New technologies that provide for higher genotyping resolution will be needed to facilitate and improve these studies going forward. Amor et al. (2020*) describe one such innovation in next generation sequencing. Better insights into clonal plant population ecology and evolution, more *in situ* data at high resolution and across species, and novel approaches to concepts such as fitness, fecundity, heterozygosity, and diversity are required to improve upon standard ecological and evolutionary models developed with marginal consideration for clonal plants (but see Winkler et al., 1999; Tuomi and Vuorisalo, 1989).

3.2. How does a clonal plant allocate resources?

3.2.1. What we know

Most clonal plants have two modes of reproduction: they produce sexual offspring via seeds and asexual offspring by vegetative growth. As both reproductive modes have their costs, we can expect a tradeoff of investments between sexual and asexual reproduction, although experimental results on intraspecific variability are equivocal (Chaloupecká and Lepš, 2004; Coelho et al., 2005). One comparative study of seed production per area in nearly 500 species concluded that clonal plants have lower seed production than non-clonal plants and that the difference is especially pronounced in clonal plants with long rhizomes (Herben et al., 2015).

Most of the organs of clonal growth are positioned belowground and often function in storage. Building extensive belowground structures that are not directly involved in resource acquisition but enable clonal growth and regeneration after damage or seasonal rest incurs costs. However, these costs have not been fully studied, and investments into clonal growth versus investment into regeneration ability are rarely separated. For example, in fire-prone areas, so-called resprouters relying on regeneration from belowground organs after fire have lower competitive ability and seed production compared to species that rely on seed regeneration (Midgley, 1996; Bellingham and Sparrow, 2000). Similarly, perennial herbs of temperate regions invest more in belowground biomass than annual plants due to storage of carbohydrates and building of bud banks for regeneration after seasonal rest. This calls for a more detailed analysis of costs due solely to resource storage, which is common also in nonclonal species, and costs specifically associated with clonal reproduction.

Martínková et al. (2020*) showed that clonal herbaceous perennials invest relatively more in belowground biomass than non-clonal ones do. We also know that biomass of rhizomes, the most common clonal growth organ, in grassland communities ranges from 30 % to nearly 100 % of aboveground biomass depending on productivity and management (Klimešová et al., 2018). Similar values were obtained for single clonal plant genets in experiments (Fiala, 1978; Jitka Klimešová, unpublished data). The comparison of clonal and non-clonal perennial herbs and their investments can provide useful information when clonality is provided by belowground organs. However, quantification of investments into aboveground clonal growth organs like stolons that do not have a regeneration function is not straightforward, partly because they are green and obtain at least part of their necessary resources themselves.

Different clonal growth organs clearly have different costs to produce, and we can speculate that these costs, apart from usually considered functions of the organs, might be partly responsible for their distribution along environmental gradients. For example, long, hypogeogenous rhizomes are more common in wetter and more nutrient-rich soils. This might be at least partly due to the large costs of these rhizomes (Craine et al., 2001).

3.2.2. Research needs

To obtain a more complete understanding of clonal growth strategies, quantifying the investments into clonal growth organs is necessary and must cover not only resource allocation in individuals but also consequences for dynamics of population subject to different perturbations or situations. For example, low seed production in clonal plants may be compensated by clonal regeneration in an established population (Herben et al., 2012) but might result in failure to establish new populations. To obtain reliable and comparable data for different clonal species despite their high morphological diversity, a standardized method for measuring investments was proposed by Goldberg et al. (2020*) for plants with hypogeogenous rhizomes, one of the most common clonal growth organs. Methods for evaluating the investments in other types of clonal growth organs are still waiting to be developed. These methods will allow us to understand only a part of the story about resource allocation concerning clonal growth but will represent an important step forward. Finally, clonal growth, in particular that based on belowground organs, should be recognized as a general strategy of resource storage, an important dimension of plant life often overlooked (but see Iwasa and Kubo, 1997; Fischer et al., 2011; Klimešová et al., 2018).

3.3. How do clonal plants respond to disturbance?

3.3.1. What we know

Disturbances are contextualized by two key elements: disturbance to the plant itself (removal or destruction of plant biomass; Grime, 1979) and disturbance to the environment of the plant (changes in competitors, nutrient pools, etc.; Battisti et al., 2016). Both of these aspects likely lead to the evolution of clonal plant traits and to allocation to vegetative reproduction over sexual reproduction (Klimešová et al., 2017). Interaction between clonal plants and disturbance has long been suggested (Sebens and Thorne, 1985), and change to the disturbance regime is a strong factor controlling clonal plant response (Herben et al., 2018). Bellingham and Sparrow (2000) suggested an advantage of asexual versus sexual reproduction under intermediate frequency and low intensity of disturbance. Typical heterogeneity of landscapes likely offers advantage to both reproductive modes.

There are four responses or behaviors that clonal plants are especially capable of, compared to non-clonal organisms: (1) resource sharing among physiologically integrated ramets, (2) ability to track resources through foraging behavior, (3) short-range dispersal using competitively superior ramets, and (4) longevity enhancing genet persistence (Svensson et al., 2013). Such traits confer advantages of clonal plants over non-clonal plants with respect to physical disturbances to the plant itself such as herbivory (Bittebiere et al., 2020*) and flooding (Martínková et al., 2020*) and to alterations to the environment due to disturbance, such as heterogeneity (Franklin et al., 2020*). However, it is unclear if these traits help following all disturbances. Clonal plants tend to be less dominant following disturbance (Klimeš et al., 1997), and clonal plant response to disturbance has known costs (Martínková et al., 2020*).

In an examination of plant traits related to disturbance regime in predominantly herbaceous flora, the traits of life span, clonal growth, and resprouting showed a stronger relationship with the environment than leaf, height, or seed traits (Herben et al., 2018). Clonal plants are expected to allocate more resources to seeds when long-distance dispersal is advantageous and when probability of establishing seeds in newly formed gaps or local ramet density are high (Janovský and Herben, 2020*;Bittebiere et al., 2020*). High frequency disturbances lead to colonization opportunities that favor long-distance dispersal and high seed production, and thus clonal plants are less dominant under more frequent disturbance (Janovský and Herben, 2020*). Indeed, on the slow-fast life history continuum (Salguero-Gómez et al., 2016), clonal plants in general are slower than non-clonal ones (Janovský and Herben, 2020*), which tends to make sense given the strategy of investing in belowground structures that allow individuals to both recover from injury (Martínková et al., 2020*) and overcome negative effects of environmental heterogeneity (Escandón et al., 2020*). However, this advantage comes with a cost, and it likely takes time to acquire this advantage (Martínková et al., 2020*). Thus, based on the life history and traits of clonal plants, we would not expect clonal plants to dominate frequently disturbed areas although they still may be present (Herben et al., 2018). What about areas of low disturbance?

In general, low disturbance means increased competition or harsh abiotic environments (Grime, 1977); the latter are not discussed here. Bittebiere et al. (2020*) provide an excellent review of clonal plants and competition in which they elucidate that clonal types differ in their competitive ability and that competition is mostly among connected parts of a clone (i.e., fragments), not disconnected ramets or genets. Phalanx types, clumping types in which few or short internodes result in closely packed ramets (Lovett-Doust, 1981), tend to be more common in homogenous environments than guerrilla types, spreading types with many or long internodes resulting in widely spaced ramets. Clumped growth structure may help competitive ability by decreasing interspecific competition and using size as a type of priority effect. Larger individuals exploit more resources and thus compete better, but this advantage again takes time. In a comparison of 17 congeneric pairs of clonal and non-clonal herbs, clonal individuals were generally larger than non-clonal ones (Martínková et al., 2020*). Guerilla types tend to be more rapid colonizers and to be found in heterogenous environments, and are able to take advantage of their disjunct resources through integration (Franklin et al., 2020*). However, guerilla types are poorer competitors than phalanx types in the colonization-competition trade-off. Increased competition can lead to either increased sexual reproduction or increased allocation to reserve organs (i.e., to storage), and the latter is perhaps a bet-hedging strategy as the individual awaits some form of disturbance. The increased storage in belowground organs allows persistence and recovery (Roiloa, 2019*;Escandón et al., 2020*). Thus, as Martínková et al. (2020*) hypothesize, clonal plant dominance should be in intermediately disturbed areas; and as Janovský and Herben (2020*) suggest, the main advantage of clonality may be that it relaxes adaptative constraints of the fecundity versus longevity trade-off.

3.3.2. Research needs

Scale is always an issue when discussing disturbance. While relationships between, say, climate and clonal traits may be evident at the biogeographic scale (Chelli et al., 2020*; Vesk and Yen, 2019*), and generalities may also be found among disturbance and traits (Herben et al., 2016, 2018; Janovský and Herben, 2020*), plant response to disturbance may be on a more local scale, that of competition and integration. Studies at multiple scales could help elucidate the most meaningful traits and growth strategies for modeling vegetation dynamics and separate the direct impacts of disturbance on plants, such as herbivory, from the indirect impacts, such as increasing or decreasing heterogeneity.

Studies on congeneric pairs of clonal and non-clonal species (e.g., Martínková et al., 2020*) are needed to determine the effect size of being clonal (Vesk and Yen, 2019*), and such studies need to follow standardized protocols of data collection (Goldberg et al., 2020*). Categories of clonal plant organs have helped elucidate the differing importance of traits such as integration and storage, but we do not fully understand their relative advantages in different environments. The colonization-competition tradeoff requires further scrutiny, as we are hindered by a lack of knowledge of the relative amounts of sexual and vegetative reproduction of many species (Huebner et al., 2019*) and by conflicting results on how species respond reproductively to disturbance (Martínková et al., 2020*) and competition (Bittebiere et al., 2020*).

3.4. Does clonal growth promote invasiveness?

3.4.1. What we know

Invasiveness can be defined as propensity to spread into new places and negatively affect things already there, and is of particular ecological concern in introduced species, those intentionally or unintentionally transported outside their native range by human action (Alpert et al., 2000). Evidence to date suggests that clonal growth is associated with invasiveness in introduced plants (Cadotte et al., 2006a). For example, clonal species are overrepresented among introduced, invasive plant species in the Czech Republic (Pyšek et al., 1995), invasive woody plants in New England (Herron et al., 2007) and North America (Reichard and Hamilton, 1997), introduced species in Ontario (Cadotte et al., 2006b), invasive species in China (Liu et al., 2006), and neophytes in Germany (Küster et al., 2008). Worldwide, clonal introduced species reduce richness of natives more than non-clonal introduced species worldwide (Vilà et al., 2015).

The association between invasiveness and clonality does appear to have limits. For instance, it is confined in some cases to specific habitats within a region, such as those disturbed by humans (Lloret et al., 2005; Holmes and Matlack, 2019), and studies of some regions have found no association (Gassó et al., 2009; Tecco et al., 2010). Moreover, no study appears to have systematically and quantitatively shown that clonality explains more than a small proportion of the invasiveness of introduced plant species (e.g., Pyšek et al., 2015).

If clonal growth significantly promotes invasiveness, which properties of clonal growth are responsible? There is little evidence so far that capacity for resource sharing between established, adult ramets, a signature property of clonal growth, is involved. Comparisons of more to less invasive, introduced, clonal species within genera have found no clear evidence that more invasive species have greater capacity for integration (Portela and Roiloa, 2017; Roiloa et al., 2019). A meta-analysis of studies on clonal integration in various species likewise failed to find that integration increased clonal performance more in more invasive species although integration did increase performance of individual ramets with relatively low access to resources more in more widespread introduced species (Song et al., 2013). Roiloa et al. (2016) similarly found that integration benefited apical ramets but not whole groups of ramets more in introduced than in native clones of *Carpobrotus edulis*. A different type of comparison, between introduced invasive species and congeneric native species, did find that clonal integration increased accumulation of mass more in the introduced species (Wang et al., 2017).

However, resource sharing in the form of support of new offspring ramets by parental ramets seems likely to contribute to invasiveness. Resource sharing from parent to offspring ramets increases their probability of survival, which could increase the spread and thus invasiveness of clones. Several studies have concluded that invasiveness increases with capacity for clonal spread (Suehs et al., 2010; Burns, 2006) and that introduced populations show more clonal growth than native populations of the same species (Jakobs et al., 2004; Douhovnikoff and Hazelton, 2014; Shah et al., 2014). Parental support could explain why tall but not short clonal species often increase in dominance when nutrient levels increase in grasslands (Dickson et al., 2014): when adult ramets overtop neighbours but juveniles do not, support facilitated by high nutrient availability allows juveniles to survive and grow up into the light. Canavan et al. (2019) found that grasses more than 2 m tall were about three times more likely to naturalize following introduction than shorter grasses, and tall clonal grasses account for much of the loss of diversity in some grasslands after long-term addition of nutrients. More generally, resource sharing may promote competitiveness when juvenile but not adult ramets are subject to asymmetric competition (Wang et al., 2021).

Reproduction is a key factor involved in the successful establishment and spread following long-distance dispersal (e.g., Moravcová et al., 2015). Asexual reproduction might be favored under unreliable conditions, providing reproductive assurance and persistence of small populations and adaptive genotypes (see review by Vallejo-Marín et al., 2010). Combined reproduction by seed and clonal growth can enable species to spread on two scales at once, distances of meters to hundreds of meters by seeds and over centimeters to meters by clonal growth, analogous to the spread of wildfire by airborne sparks that set small individual fires. For example, the invasive, clonal species *Acroptilon repens* and *Phragmites australis* can produce an advancing front of growing patches (Duncan et al., 2017; Gaskin and Littlefield, 2017).

In aquatic habitats, individual ramets or small groups of connected ramets fragmented from larger groups by disturbance may serve the dispersal function of seeds (Vilas et al., 2017; Uya et al., 2018). In two highly invasive, aquatic clonal species provisioning of ramets by resource sharing before fragmentation greatly increased their subsequent growth (Dong et al., 2019a*; Adomako et al., 2021). Very limited evidence suggests an association between probability of establishment of fragmented ramets and invasiveness in aquatic species: fragments of the invasive, introduced species *Elodea nutalllii* showed a higher probability of forming roots than co-occurring native species in German streams (Heidbüchel et al., 2019). Cultivated fields may be another type of habitat where fragmentation of clones leads to wide dispersal (Klimešová, 2011)

Physiological integration between connected ramets might also promote invasiveness through signaling for responses such as division of labor. For example, Roiloa et al. (2019) documented greater induction of division of labor in the more invasive of two introduced species of *Carpobrotus*, and Chao et al. (2020) reported greater effects of integration on allocation of mass, consistent with greater capacity of division of labor, in introduced than in native genotypes of *Hydrocotyle vulgaris*.

One obvious consequence of clonal growth is that clonal offspring are genetically identical. This could preclude post-introduction adaptation, although epigenetic variation might partly compensate (e.g., Guarino et al., 2019; Shi et al., 2019). On the other hand, clonal growth could perpetuate particularly invasive genotypes. Populations of introduced, invasive, clonal species in a region sometimes consist of a single clone (e.g., Canavan et al., 2017). This might just be because only one clone has been introduced and conditions permit clonal but not sexual reproduction (Ferrero et al., 2020). Alternatively, some invasive clones may have exceptionally high fitness in their introduced range. For instance, numerous clones of the highly invasive species *Arundo donax* (Canavan et al., 2017) and *Eichhornia crassipes* (Zhang et al., 2010) have been introduced but one clone clearly dominates the introduced range of each worldwide.

Selective placement of offspring is another potential advantage of clonal growth. Chen et al. (2019) recorded that invasive clonal species grew fewer but larger ramets in low-nutrient patches than native, clonal congeners. A novel study by Reijers et al. (2019) concludes that spread of the clonal grasses *Ammophila arenaria* and *A. breviligulata* on coastal sand dunes can be modeled as a random walk with high variability. The spatial pattern of spread of the former species, which builds higher dunes and has been more invasive, traps more sand per unit of clonal growth.

Besides testing for links between invasiveness and individual, special properties of clonal growth, researchers have asked if clonality is associated with properties linked to invasiveness in plant species in general. Many invasive species show relatively great increases in growth in response to high nutrient availability (Alpert et al., 2000). Wang et al. (2019*) report that a set of invasive clonal species responded more to high nitrogen than co-occurring native clonal species. In a rare study of evolution of clonal traits following introduction, Bock et al. (2018) conclude that the positive effect of high water availability on the clonal spread of *Helianthus tuberosus* increased following introduction.

High competitive ability is another obvious mechanism of spread in native communities after introduction. Clones of *Phragmites australis* introduced from Europe have displaced native clones in parts of North America, and Pyšek et al. (2020) show that European clones in both Europe and North America have greater competitive ability than native North American clones. Competitiveness and response to nutrients could combine to promote invasiveness, but recent studies have variously reported that higher nutrient levels increased (Liu et al., 2019; Liang et al., 2020) or decreased (Wang et al., 2019*) the competitive ability of invasive clonal species.

3.4.2. Research needs

Investigations of the link between clonal growth and invasiveness in plants so far suggest that clonal growth is among the plant traits that promote invasiveness. More research is needed on two main fronts. First, we need to know more about how habitat and introduction history may interact with clonal traits to make species invasive following introduction. For example, do certain disturbance regimes favor invasion by clones?

Second, we need to better understand which traits of clonal growth confer invasiveness. Comparisons between more and less invasive, introduced clonal species or between more and less invasive genotypes within species seem especially promising. In particular, demographic, morphological, and physiological comparisons between the genotypes that dominate very widespread invasion by some clone species and other introduced genotypes in the same species could tell us if these dominant clones share common features that make fitness remarkably high. Comparison between introduced and source populations of invasive clonal species could show whether introduced populations have evolved greater propensity for clonal growth and suggest if this has traded off with reduced sexual reproduction.

More studies of interactive effects of clonal and other traits may reveal effects of clonality on invasiveness not apparent when clonal species are considered as a whole. The prime case of this so far seems to be the spread of clonal species that are also relatively tall and that place new ramets relatively far from parents within grasslands following increase in soil nutrient availability. Finally, few studies have documented or modeled the combined effects of fragmentation and clonal growth on the spread of introduced species.

3.5. How does clonal multiplication affect our understanding of plant demography?

3.5.1. What we know

Demographic approaches of clonal plants provide an ideal platform on which to identify the role that clonality as an ecological strategy plays in the life cycle of plants, their viability, and thus the selective forces that underlie evolution of clonality and its loss (Klimeš et al., 1997; Salguero-Gómez, 2018; Janovský and Herben, 2020*). Indeed, population ecology must inevitably deal with the hierarchical nature of individuality in clonal plants, ranging from ramets through clonal fragments (i.e., connected and potentially physiologically integrated sets of ramets) to genets. From a demographic perspective, clonal growth can be considered as a type of reproduction during which new independent units arise; from the evolutionary perspective, clonality is a form of individual growth that enlarges the genet in space and consequently in time (Aarssen, 2008) through ramet turnover. Indeed, clonal species seem to have traded the postponing of ramet senescence for the postponing of genet senescence (Salguero-Gómez, 2018), decelerating recombination rates and evolutionary changes as a side effect (Bousquet et al., 1992; Smith and Donoghue, 2008; Orive et al., 2017). However, the relationship between evolutionary rates and zygote formation may be blurred by somatic mutations taking place in the extensive vegetative tissues of clonal plants (Schoen and Schultz, 2019).

Demographic analyses have shown that clonal growth can act as an effective means of maintaining diversity in disturbed ecosystems. For instance, the turnover of ramets in genets enables clonal species to occupy habitats with stronger disturbance regime than permitted by sexual reproduction only (Klimeš et al., 1997; Herben et al., 2018; Janovský and Herben, 2020*). Clonal growth also increases variation in ramet population growth rates of clonal species, permitting the species to exploit "windows of opportunity" under favourable conditions (Janovský and Herben, 2021*). Moreover, clonal growth seems to relax to some extent the demographic constraints imposed by a plant's developmental and architectural constraints and allow plants to evolve different life history strategies than those available to non-clonal plants. For instance, with short-lived ramets and clonal growth it is possible to maintain both short and long genet generation times, while in non-clonal species ramet and genet life span cannot be decoupled (Eriksson and Bremer, 1993; Salguero-Gómez, 2018).

Besides the trade-off between clonal growth and sexual reproduction, clonal growth can affect plant species' demography through further decreases of reproduction by seeds due to pollen limitation (Barrett, 2015), especially in self-incompatible clonal species (Honnay and Jacquemyn, 2008). This is mainly due to production of new ramets increasing the abundance of sources of self-pollen in a ramet's surroundings and thus possibly affecting the quality of generative offspring (Harder and Barrett, 1995; Vallejo-Marín et al., 2010) through increased risk of self-pollination (e.g. Hu et al., 2015). With the recent loss of pollinator species worldwide (Potts et al., 2010), the ability of asexual clonal plants to circumvent pollinator needs deserves scrutiny.

3.5.2. *Research needs*

The existing knowledge of demographic characteristics of ramets is relatively good, though for a rather limited number of clonal plant species (e.g. Eriksson, 1988; Carlsson and Callaghan, 1990; Jongejans and de Kroon, 2005; Černá and Münzbergová, 2013). This knowledge gap is particularly obvious when considering that two thirds of the Plant Kingdom has clonal abilities (Herben and Klimešová, 2020). Of greater concern is that similar information about genets is even rarer, which precludes general evolutionary comparisons of demography of non-clonal and clonal plants (Janovský et al., 2017). While studies on genet demography do exist (Eriksson and Bremer, 1993; Fair et al., 1999; Araki and Ohara, 2008; Czarnecka, 2008; de Witte et al., 2011; Matsuo et al., 2018), they rarely provide sufficient detail to integrate ramet within genet dynamics. The much-needed integration of both biological levels of organisation in clonal plants would allow for a much deeper understanding of the evolutionary pressures, ecological strategies, and conservation biology outcomes of this prominent plant trait.

A possible solution for this gap could be to derive genet demographic characteristics by simulating genet life histories from the existing demographic models of populations of ramets (Janovský et al., unpublished). However, such an approach cannot be applied to species with strong integration of ramets within the genet because of differing vital rates for ramets of different types (e.g. Carlsson and Callaghan, 1990; Münzbergová et al., 2005; Wikberg and Svensson, 2006), and whole genets need to be followed in a way that does not damage them and so jeopardise the long-term monitoring necessary in most demographic studies of plants. Demography of whole genets thus seems to be an obvious choice for compact tussock-forming species and may work well in the short term; however, clear delimitation of working units may become problematic as clonal fragments disintegrate over longer time periods. For such species, we therefore need more studies coupling demography with genetic techniques (such as Harada et al., 1997; Amor et al., 2020*; Ricono et al., 2020*; Tsujimoto et al., 2020).

The way plant demographers define individuals of clonal plants

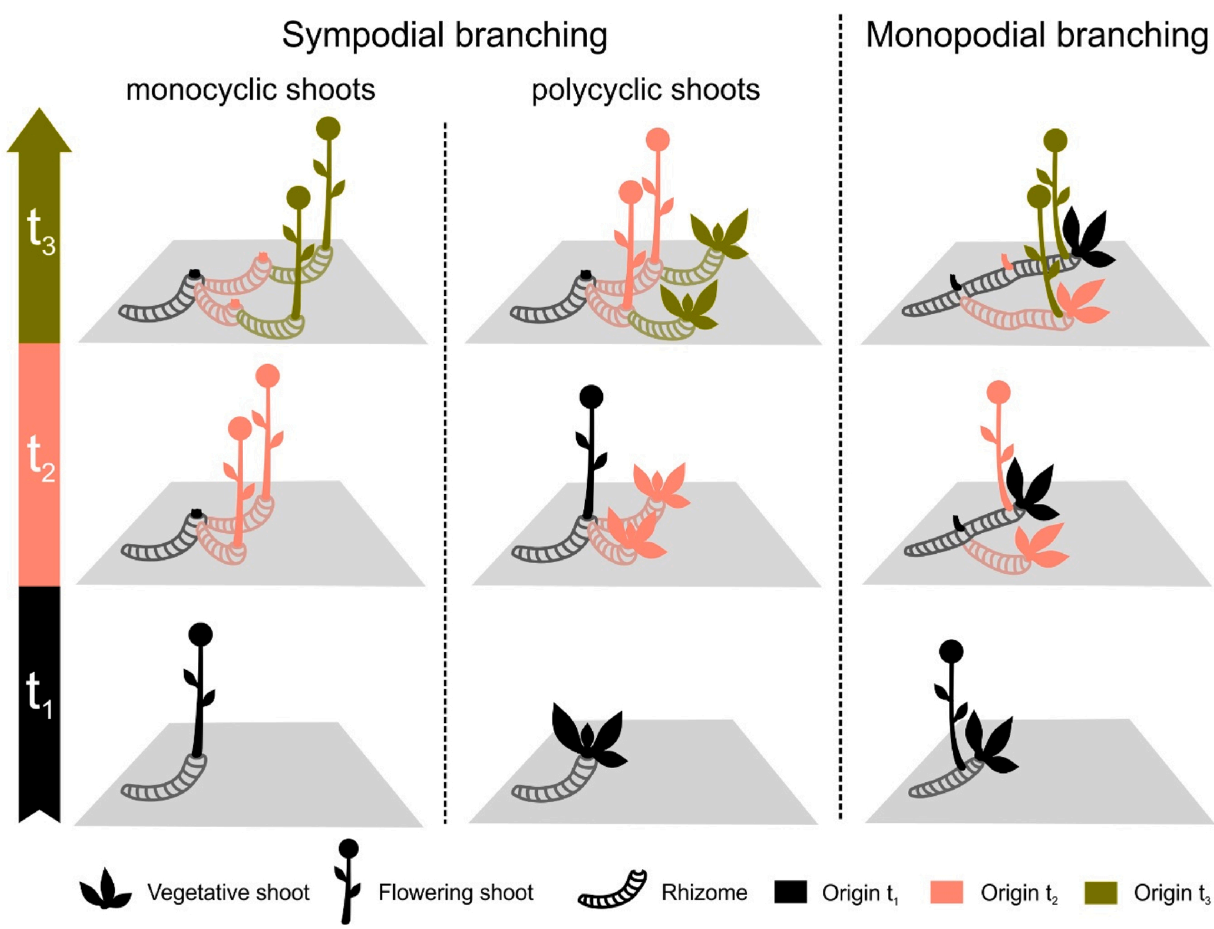

Fig. 1. Architectural development of sympodial and monopodial branching over three time periods (t). Demographers frequently fail to record clonality in species with monocyclic shoots, whose ramets emerge at different positions each season. There is also a lack of studies on species with very low and very high intensities of clonal growth, especially those with extensive rhizome systems. Illustration by Alena Bartušková.

(ramets or genets) to be observed and modeled largely depends on plant species' architecture, mainly type of branching and shoot cyclicity. Plant architecture also affects the probability of recording clonal growth; e.g., demographers more frequently fail to record clonality in species with monocyclic shoots (see Serebriakova, 1977, for definition) where ramets emerge at different positions each season (Fig. 1). There is a general lack of studies on species with very low (due to frequent omissions of recording clonal growth) and very high intensities of clonal growth (due to difficulties in tracking ramet identity in such species; Janovský et al., 2017). This especially holds true for species possessing extensive rhizome systems and featuring a rich bud bank such as most grasses (e.g. Mortimer, 1983; Ott and Hartnett, 2015). In general, we need to differentiate among clonal species more according to their architecture in future studies reflecting thus the multitude of events (and possibly selection pressures) in which clonality has evolved, as well as sexual/asexual ratios and changes to those ratios resulting from environmental factors.

3.6. Does clonality affect community assembly?

3.6.1. What we know

In 1995, a review by Oborny and Bartha (1995) reflected on the fact that clonal reproduction seemed to be a prevalent process in many plant communities but had not been systematically examined as a process structuring communities. Similarly, they posited that differences in clonal traits (not yet denoted by this term then) are likely to be important for driving species coexistence. Since then, a number of studies on the topic have been published, but these existing papers are more a haphazard collection of separate studies than a systematic foray into a single phenomenon.

Clonality has generally been an overlooked trait in studies of community assembly. While it has been argued that trait space is incomplete without traits on resource storage, clonality, and resprouting (Ottaviani et al., 2017; Klimešová, 2018; Klimešová et al., 2018), most trait-based studies on community assembly have failed to take these traits into account. Important to note, clonality is not a single trait, in spite of references to a non-clonal versus clonal dichotomy in many papers. Plants reproducing clonally may have long or short inter-ramet distances, with very different bearing on species interactions, coexistence, and resulting community assembly. There may be less difference between a non-clonal plant and a plant with a slowly growing, epigeogenous rhizome than between two clonal plants that differ strongly in inter-ramet distances and multiplication rates. It is worth noting here that these traits have generally low phylogenetic conservatism (Herben and Klimešová, 2020), implying that their variation cannot be easily proxied by phylogenetic data and must be approached directly.

The few existing studies on clonal traits have almost always found signals of convergence in traits of clonal growth, often stronger than convergence in leaf, height, and seed traits such as height at maturity, specific leaf area, or seed mass (de Bello et al., 2011; Ye et al., 2014, Vojtkó et al., 2018;Chelli et al., 2020*), which has generally been interpreted as environmental filtering. It is worth noting that most of these studies with the exception of Ye et al. (2014) have collected data on a fairly narrow range of community types, indicating strong environmental effects. On the other hand, most studies have also found that clonal and non-clonal species coexist in almost all community types and that variation in clonal trait values within a community can still be large (de Bello et al., 2011; Vojtkó et al., 2017). Demographic differentiation of coexisting species involves an important axis of clonal dispersal even at the small scale (Herben et al., 2019). Longitudinal data also show that it does not need to change substantially through time (Duchoslavová and Herben, 2020*). This phenomenon is too common to be due to simple sampling phenomenon from a larger species pool (de Bello et al., 2011;Vojtkó et al., 2018), indicating that differences in clonal growth are one of the patterns in multispecies communities and elucidating its potential role in species assembly.

However, analysis of community traits structure has its obvious limitations; namely, very different mechanisms may produce indistinguishable trait patterns (Mayfield and Levine, 2010; Kraft et al., 2015; Münkemüller et al., 2020). In particular, trait convergence as a putative signature of environmental filtering is difficult to distinguish from resource competition (Kraft et al., 2015). It is therefore essential to link functional traits with recognized assembly mechanisms (Adler et al., 2007), namely differences in fitness and stability.

Current data show that differences in clonality per se are unlikely to contribute to fitness differences. Clonality is, as a rule, associated with large belowground structures used for resource storage and resprouting. Building storage clearly constrains investment into aboveground organs, which play the key role in asymmetric competition for light (Suzuki and Hara, 2001; Martínková et al., 2020*). This is likely to make competition more symmetric and thus reduce fitness differences among species; as a result, niche differences necessary for coexistence should be weaker. However, this is not restricted to clonal plants only, as all species in habitats where large belowground storage is necessary for survival (e.g., due to frequent, unpredictable disturbance) show similar patterns. Fitness effects of clonality may change with environmental conditions: clonality interacts with plant height for fitness response to fertilization (Eilts et al., 2011; Gough et al., 2012), and the role of clonal integration for coexistence differs among habitats (Pennings and Callaway, 2000).

It is less obvious whether and how clonality can affect niche differences. Here it is important to note that the role of clonality in coexistence is intimately linked with spatial processes (Oborny and Bartha, 1995; Gigon and Leutert, 1996) which to an important degree modify interactions predicted by nonspatial models (Bolker et al., 2003). Clonal plants tend to position their offspring close to their parental individuals, resulting in strong spatial structure that constrains, often in a nontrivial way, how individual species can coexist (see e.g. Stoll and Prati, 2001; Lenssen et al., 2005 for experimental demonstrations). In principle, there are three important mechanisms by which clonality may affect species coexistence through spatial processes. First, founder-controlled communities in spatially explicit systems can, in contrast to nonspatial systems, remain diverse in spite of potential competitive exclusion of one species by the other, essentially due to an equalizing mechanism that prevents fitness differences from prevailing (Molofsky et al., 2002; Bolker et al., 2003). Although this mechanism does not account for stable-state coexistence (Chesson and Huntly, 1997), time to extinction is far too long in realistic settings, making it effectively a mechanism of coexistence. Second, clonality may strongly contribute to fast occupancy of space in open patches in the community by a process that has been termed spatial successional niches (Bolker et al., 2003; see e.g. Vojtkó et al., 2017, for an experimental demonstration). This process would take place only in moderately disturbed systems; differences in traits of spatial spreading do not generate niche differentiation in model systems with no disturbance. Third, spatial dynamics due to clonal growth may be linked with negative plant-soil feedback (Bever, 2003; Bever et al., 2015).

3.6.2. Research needs

Most important, we need to know how clonality and different values of clonal traits contribute to the key processes of species coexistence, namely niche differences and differences in fitness, taking into account the multidimensionality of clonal traits. At the same time, we need to know relationships between clonal traits and other sets of traits, namely those known to affect species interactions, such as plant height and root foraging, as well as the relationship between clonality and growth form. Existing data support the contention that woody species are less clonal and show morphologically and functionally different types of clonality. We need to be sure that this pattern is general; if it is, it can be indicative of the role of clonality both in environmental response and interactions with neighbours. We also need to know more about the relationship of clonality and clonal traits to regeneration strategies; while it is commonly assumed that regeneration is associated with clonality

(Bellingham and Sparrow, 2000), many non-clonal species are able to regenerate, and the link between clonality and disturbance is weaker than often assumed (Herben et al., 2018; Martínková et al., 2020*).

Phylogenetic analyses have shown that clonality is a trait that is fairly easily acquired and lost; this indicates that individual species respond to selective forces in favor of and against clonality (Herben and Klimešová, 2020). While these forces may not necessarily be due to coexistence-related processes (e.g., they may be due to a link between clonality with sexual reproduction), we clearly need to know the role of environmental effects, both abiotic and of neighbours. This can hardly be attained without comparative experimental approaches, examining response to competition of sets of species differing in their clonal traits.

4. Conclusion

This introductory overview and this special issue highlight both the obstacles and opportunities for incorporating clonal plant knowledge into mainstream plant ecology and evolution studies. Clonal plants take modularity to another level, offering both novel traits and responses to a variety of environmental biotic and abiotic factors. This seemingly easy to gain or lose strategy is related to climate, disturbance, species interactions, community assembly, and ecosystem function. Including clonality will increase predictive power and improve models, albeit not without challenges.

One major challenge includes the continued lack of tools that capture the hierarchical nature of an individual in clonally growing plants, so we need studies utilizing more genetic demographic information and studies at the whole genet scale. Studies have concentrated on relatively few species, and studies on species with very high intensities of clonal growth are largely lacking. A second major challenge is lack of a standardized way of measuring allocation in various clonal types and of comparing resulting architectures. Despite these challenges, similarity of questions and importance of clonal plants to overall biomass and diversity of ecosystems implies an easy incorporation into mainstream plant studies. Currently, we possess a number of novel tools, ranging from large trait databases through experimental approaches such as congeneric pair studies to phylogenetic and molecular techniques, to address the impact of clonality on community assembly, biological invasions, and the responses of plants to environmental factors. With land use impacting communities and ecosystems as much as climate change, we need to link colonization and resilience traits to disturbance regimes and to scale up to communities and ecosystems.

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